Implicit Differentiation II

Revisit: Comparing Explicit vs. Implicit Differentiation

- **Question:** Let's revisit $xy + y = e^x$. What is $\frac{dy}{dx}$? Check that it's the same as what you got previously!

Method 1: Explicit Differentiation (from previous lecture)
Solve for y explicitly first: $y(x+1) = e^x \implies y = \frac{e^x}{x+1}$.
Differentiating via the Quotient Rule:

$$\frac{dy}{dx} = \frac{(x+1)\frac{d}{dx}[e^x] - e^x\frac{d}{dx}[x+1]}{(x+1)^2} = \frac{(x+1)e^x - e^x(1)}{(x+1)^2} = \frac{xe^x + e^x - e^x}{(x+1)^2} = \frac{xe^x}{(x+1)^2}.$$

Method 2: Implicit Differentiation (new approach)
Treat y as an implicit function of x, writing $y = y(x)$:

$$x \cdot y(x) + y(x) = e^x.$$

Differentiate both sides with respect to x using the **Product Rule** on $x \cdot y(x)$:

$$\underbrace{\frac{d}{dx}[x] \cdot y(x) + x \cdot \frac{d}{dx}[y(x)]}_{\text{Product Rule on } x \cdot y} + \frac{d}{dx}[y(x)] = \frac{d}{dx}[e^x]$$

$$1 \cdot y(x) + x \cdot y'(x) + y'(x) = e^x.$$

Factor out $y'(x)$ to isolate the derivative:

$$y'(x)(x+1) + y(x) = e^x \implies y'(x)(x+1) = e^x - y(x) \implies y'(x) = \frac{e^x - y(x)}{x+1}.$$

Verification: Substitute the original relation $y(x) = \frac{e^x}{x+1}$ into our implicit derivative formula:

$$y'(x) = \frac{e^x - \frac{e^x}{x+1}}{x+1} \cdot \frac{x+1}{x+1} = \frac{e^x(x+1) - e^x}{(x+1)^2} = \frac{xe^x + e^x - e^x}{(x+1)^2} = \frac{xe^x}{(x+1)^2}. \quad \checkmark \text{ (Identical!)}$$

The Core Mechanism of Implicit Differentiation

We pretend y is a function of x, making expressions like y^2, y^3, e^y, etc. into compositions $[y(x)]^2, [y(x)]^3, e^{y(x)}$. We then apply the **Chain Rule** and **Product Rule**:

- $\frac{d}{dx}[y^n] = ny^{n-1} \cdot \frac{dy}{dx}$ (e.g. $\frac{d}{dx}[y^3] = 3y^2 \cdot y'$)
- $\frac{d}{dx}[x \cdot y] = 1 \cdot y + x \cdot y'$ (**Product Rule**)
- Gather all terms with y' on one side, factor out y', and divide by the coefficient expression.

Implicit Differentiation II — Practice Questions

Question 1: Finding Tangent Lines to Algebraic Curves

1. (a) Find the tangent line to $x^2 + xy - y^3 = xy^2$ at the point $(1, 1)$.

Deriv. of $[y(x)]^3$:
- inner: $y(x)$, outer: u^3
- inner′: $y'(x)$, outer′: $3u^2$
- $\implies 3y^2 \cdot y'$

Deriv. of $[y(x)]^2$:
- inner: $y(x)$, outer: u^2
- inner′: $y'(x)$, outer′: $2u$
- $\implies 2y \cdot y'$

Step 1: Write y explicitly as $y(x)$ and differentiate:

$$x^2 + x \cdot y(x) - [y(x)]^3 = x \cdot [y(x)]^2$$

Differentiating each term with respect to x:

$$2x + \big(x \cdot y'(x) + y(x) \cdot 1\big) - 3[y(x)]^2 y'(x) = x \cdot \big(2y(x)y'(x)\big) + 1 \cdot [y(x)]^2$$

Step 2: Collect and isolate y' terms:

$$2x + xy' + y - 3y^2y' = 2xyy' + y^2$$

$$xy' - 3y^2y' - 2xyy' = y^2 - 2x - y$$

$$y'(x - 3y^2 - 2xy) = y^2 - 2x - y$$

$$\frac{dy}{dx} = \frac{y^2 - 2x - y}{x - 3y^2 - 2xy}$$

Step 3: Evaluate slope at $(x, y) = (1, 1)$:

$$m = \left.\frac{dy}{dx}\right|_{(1,1)} = \frac{(1)^2 - 2(1) - 1}{1 - 3(1)^2 - 2(1)(1)} = \frac{1 - 2 - 1}{1 - 3 - 2} = \frac{-2}{-4} = \frac{1}{2}.$$

Step 4: Tangent Line Equation at $(1, 1)$:

$$y - y_0 = m(x - x_0) \implies y - 1 = \frac{1}{2}(x - 1) \implies y = \frac{1}{2}x + \frac{1}{2}.$$

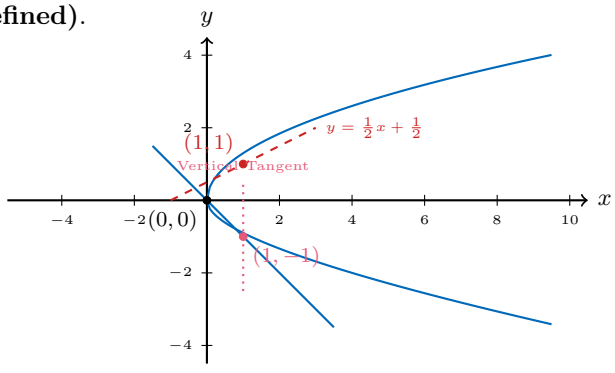
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1. (b) Here's a picture of this curve. Plug the point $(0, 0)$ and $(1, -1)$ into your expression for $\frac{dy}{dx}$. What do you get? Looking at the graph, what can you say about the slope at those points?

- **At point $(0, 0)$:**

$$\left.\frac{dy}{dx}\right|_{(0,0)} = \frac{0^2 - 2(0) - 0}{0 - 3(0)^2 - 2(0)(0)} = \frac{0}{0} \quad \textbf{(Indeterminate / Undefined).}$$

Geometric interpretation: The curve crosses itself at the origin (a node / self-intersection), meaning there are **two distinct tangent slopes** at $(0, 0)$, so $\frac{dy}{dx}$ cannot be uniquely defined single-valuedly.



- **At point $(1, -1)$:**

Denominator: $1 - 3(-1)^2 - 2(1)(-1) = 1 - 3 + 2 = 0.$

When the denominator of $\frac{dy}{dx}$ equals zero (while the numerator is non-zero or approaching limits), the tangent line is **vertical** ($\frac{dy}{dx} \to \pm\infty$).